

9(a)

any **nine** from:

- 1 hypothalamus / osmoreceptors, detects changes in water potential of blood ;
- 2 (causes) posterior pituitary gland ;
- 3 (to) release ADH into blood ;
- 4 ADH binds to receptors ;
- 5 on cell surface membrane of collecting duct (cells) ;
- 6 stimulates enzyme cascade / phosphorylase enzyme produced ;
- 7 vesicles, move towards / fuse with, cell surface membrane ;
- 8 (vesicles have) aquaporins ;
- 9 collecting duct, membranes / cells / walls, more permeable to water ;
- 10 water moves, down water potential gradient / by osmosis ;
- 11 into, (collecting duct) cells / tissue fluid / blood ;
- 12 water potential (of blood) returns to set point ;

Question	Answer	Marks
9(b)	any six from: 1 dendrites attached to cell body ; 2 nucleus in, cell body / soma ; 3 many mitochondria ; 4 much rough endoplasmic reticulum / Nissl's granules (in cell body) ; 5 long axon ; 6 synaptic knobs / synaptic bulbs / terminal branches / axon terminals ; 7 Schwann cells / myelin sheath / myelinated ; 8 nodes of Ranvier ; 9 cell body in, CNS / brain and spinal cord ; <i>accept from labelled diagram</i> <i>if sensory neurone described – max 5</i>	6